Probability course: problems and solutions

Jack Ceroni*

(Dated: Sunday 15th March, 2026)

Contents

I. Introduction 1

II. Chapter 6 (Basic Probability) 1

III. Chapter 7 (Weak Convergence) 4

IV. Chapter 10 (Conditional expectation) 4

V. Chapter 11 (Discrete martingales) 4

I. Introduction

The goal of these notes is to solve the majority of the problems from these lecture notes on basic probability and stochastic calculus, written by Gordan Žitković (beginning in Chapter 6). Here are the current tallies

- Chapter 6: 12/18
- Chapter 7: 1/8

II. Chapter 6 (Basic Probability)

Solution II.1 (Problem 6.1)**.** Recall that by definition,

$$\mathbb{E}[g(X)] = \int_\Omega (g \circ X)\ dP \tag{1}$$

In the case that g is a non-negative simple function with standard representation $g = \sum_{j=1}^N a_j \chi_{E_j}$, then we can write $g \circ X = \sum_{j=1}^N a_j \chi_{X^{-1}(E_j)}$ so that

$$\int_{\mathbb{R}} g\ d\mu_X = \sum_{j=1}^N a_j P(X^{-1}(E_j)) = \int_\Omega g \circ X\ dP \tag{2}$$

Assuming g is non-negative, we can choose a sequence of simple functions $\phi_n \to g$ a.s. with $\phi_n \leq \phi_{n+1}$. In this case, $\phi_n \circ X \to g \circ X$ from below as well, and by monotone convergence,

$$\int_{\mathbb{R}} g\ d\mu_X = \lim_{n\to\infty} \int_{\mathbb{R}} \phi_n\ d\mu_X = \lim_{n\to\infty} \int_\Omega (\phi_n \circ X)\ dP = \int_\Omega (g \circ X)\ dP \tag{3}$$

* jceroni@uchicago.edu

as desired. In the case that g is not assumed to be non-negative, then note that applying Eq. (2) to the positive and negative parts of g demonstrates that g is integrable if and only if $g \circ X$ is, and moreover that the desired integral formula holds in this case. Note that g is measurable if and only if $g \circ X$ by definition of the σ-algebras on the two respective domains.

Solution II.2 (Problem 6.2)**.** There are two parts:

1. Clearly,

$$\mu_{X_k}(B) = P(X_k^{-1}(B)) = P(\boldsymbol{X}^{-1}(\mathbb{R} \times \cdots \times B \times \cdots \times \mathbb{R})) = \mu_{\boldsymbol{X}}(\mathbb{R} \times \cdots \times B \times \cdots \times \mathbb{R}) \tag{4}$$

   Thus, if $\mu_{\boldsymbol{X}} \ll \lambda_{\mathbb{R}^n}$, then for B such that $\lambda(B) = 0$, then $\mu_{\boldsymbol{X}}(\mathbb{R} \times \cdots \times B \times \cdots \times \mathbb{R}) = 0$, so $\mu_{X_k}(B) = 0$ as well, and we have $\mu_{X_k} \ll \lambda$, as desired. To obtain the formula for the pdf, simply note that by Fubini's theorem:

$$\mu_{X_k}(B) = \mu_{\boldsymbol{X}}(\mathbb{R} \times \cdots \times B \times \cdots \times \mathbb{R}) = \int_{\mathbb{R} \times \cdots \times B \times \cdots \times \mathbb{R}} f \, d\lambda_{\mathbb{R}^n} = \int_B f_{X_k} \, d\lambda \tag{5}$$

   so by uniqueness of the Radon-Nikodym derivative, the pdf must be given by f_{X_k}.

2. Note that the diagonal $\Gamma \subset \mathbb{R}^2$ has Lebesgue measure-0. However, $(X, X)^{-1}(\Gamma) = \Omega$, and $P(\Omega) = 1$, so it is not the case that $\mu_{(X,X)} \ll \lambda_{\mathbb{R}^2}$.

Solution II.3 (Problem 6.4)**.** Obviously if X and Y are discrete, then we have B_X and B_Y with $\mathbb{P}[X^{-1}(B_X)] = \mathbb{P}[Y^{-1}(B_Y)] = 1$. Note that $\mathbb{P}[X^{-1}(B_X) \cap Y^{-1}(B_Y)] = 1$ as well. If $X(p) \in B_X$ and $Y(p) \in B_Y$, then $X(p) + Y(p) \in B_X + B_Y$, so $X^{-1}(B_X) \cap Y^{-1}(B_Y) \subset (X+Y)^{-1}(B_X + B_Y)$, hence the latter set has measure-1. Clearly, $B_X + B_Y$ is a countable collection of points, therefore Borel measurable. For the case of absolutely continuous random variables, pick some absolutely continuous X and note that $-X$ is also absolutely continuous, yet $Y = X + (-X) = 0$ satisfies $\mathbb{P}[Y^{-1}(\{0\})] = 1$ while $\lambda(\{0\}) = 0$.

Solution II.4 (Problem 6.5)**.** There are two parts:

1. Suppose we take two measurable sets A and B such that $\mathbb{P}[A \cap B] \neq \mathbb{P}[A]\mathbb{P}[B]$. If we let C be some measure-0 set, then $A \cap B \cap C$ is measure-0 as well. Hence, if we take our families to be $\{A\}, \{B\}, \{C\}$, we get the desired example.

2. In this case, it is clear that any of the cases with $k < n$ reduces to a case of $k = n$.

Solution II.5 (Problem 6.6)**.** This follows immediately from the definition of a σ-algebra generated by a random variable.

Solution II.6 (Problem 6.7)**.** Note that any intersection of a subcollection of events drawn from each of the families can be written as $(A_{i_1} \cap \cdots \cap A_{i_k}) \cap (A_{j_1}^C \cap \cdots \cap A_{j_\ell}^C)$. If $\ell = 1$, then we immediately note that

$$\begin{aligned} \mathbb{P}[A_{i_1} \cap \cdots \cap A_{i_k} \cap A_{j_1}^C] &= \mathbb{P}[A_{i_1} \cap \cdots \cap A_{i_k}] - \mathbb{P}[A_{i_1} \cap \cdots \cap A_{i_k} \cap A_{j_1}] && (6) \\ &= \mathbb{P}[A_{i_1}] \cdots \mathbb{P}[A_{i_k}](1 - \mathbb{P}[A_{j_1}]) = \mathbb{P}[A_{i_1}] \cdots \mathbb{P}[A_{i_k}]\mathbb{P}[A_{j_1}^C] && (7) \end{aligned}$$

so the result holds in this case. It then follows via a simple induction on ℓ that the result holds in general.

Solution II.7 (Problem 6.8)**.** This problem has four parts, each applying the π-λ theorem to the theory of independent families events.

1. If $R_1, \ldots, R_n$ are rectangles in $\mathbb{R}$, then note that

$$\begin{aligned} \mu_{\mathbf{X}}(R_1 \times \cdots \times R_n) &= \mathbb{P}[\mathbf{X}^{-1}(R_1 \times \cdots \times R_n)] && (8) \\ &= \mathbb{P}[X_1^{-1}(R_1) \cap \cdots \cap X_n^{-1}(R_n)] && (9) \end{aligned}$$

If the variables $X_1, \ldots, X_n$ are independent then $\mathbb{P}[X_1^{-1}(R_1) \cap \cdots \cap X_n^{-1}(R_n)] = \mu_{X_1}(R_1) \cdots \mu_{X_n}(R_n)$, hence $\mu_{\mathbf{X}}$ and the product measure agree on rectangles so they agree on all Borel sets. In particular, these measures will agree on the algebra of finite disjoint unions of rectangles. Because the measure space is question is σ-finite, the premeasures on this algebra have unique extensions to the σ-algebra generated by the algebra. Conversely, if the measures agree, then $\mu_{\mathbf{X}}(R_1 \times \cdots \times R_n) = \mu_{X_1}(R_1) \cdots \mu_{X_n}(R_n)$, so the π-systems generated by $X_j^{-1}(R)$ for each j and R a rectangle are independent. It follows from Proposition 6.14 that the σ-algebras generated by $X_1, \ldots, X_n$ are also independent.

2. Each set $X_j^{-1}(-\infty, x_j]$ is in $\sigma(X_j)$, hence the assumption of independence implies the formula. Conversely, if the formula holds, then the π-system given by all $X_j^{-1}(-\infty, x_j]$ for all x_j is independent when we vary over $j = 1, 2, \ldots, n$. Hence the corresponding σ-algebras are independent. The rays will generate the Borel σ-algebra, so these σ-algebras will in fact be $\sigma(X_1), \ldots, \sigma(X_n)$, and we're done.

3. Under the assumption of independence, we can apply Part 1 and a.e. uniqueness of the Radon-Nikodym derivative. Conversely, if the Radon-Nikodym derivatives agree, then the integrals of these functions agree on all rectangles. In other words, $\mu_{\mathbf{X}}$ and $\mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$ agree on all rectangles, hence they agree on all Borel sets, and by Part 1, $X_1, \ldots, X_n$ are independent.

4. Independence implies the formula. Conversely, let $\mathcal{P}_j$ be the π-system given by taking $X_j^{-1}(x_j)$ for all $x_j \in C_j$: clearly this is, in fact, a π-system. All of these π-systems are independent, so the corresponding σ-algebras are. Note that $\sigma(\mathcal{P}_j)$ contains all of the sets in $\sigma(X_j)$, up to potentially measure-0 sets. But any measure-0 set trivially satisfies the required conditions for independence, thus $\sigma(X_1), \ldots, \sigma(X_n)$ are independent families as desired.

Solution II.8 (Problem 6.9)**.** Note that if $X_1, \ldots, X_n$ are independent, then $\mu_{\mathbf{X}} = \mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$, so we know from basic measure theory that if $\mu_{X_j} \ll \lambda_{\mathbb{R}}$ (the Borel measure) for each j, then $\mu_{\mathbf{X}} \ll \lambda_{\mathbb{R}^n}$ as desired. Conversely, if $\mu_{\mathbf{X}} \ll \lambda_{\mathbb{R}^n}$, then if $\lambda_{\mathbb{R}}(E) = 0$, we have $\lambda_{\mathbb{R}^n}(\mathbb{R} \times \cdots \times E \times \cdots \times \mathbb{R}) = 0$ and

$$\mu_{X_j}(E) = \mu_{\mathbf{X}}(\mathbb{R} \times \cdots \times E \times \cdots \times \mathbb{R}) = 0 \tag{10}$$

as desired.

Solution II.9 (Problem 6.10)**.** There are two parts:

1. Let $\mathcal{P}_i$ be the π-system which is generated by taking all finite intersections of elements drawn from $\mathcal{F}_{i1}, \ldots, \mathcal{F}_{im_i}$. Because each $\mathcal{F}_{ij}$ is a σ-algebra, a generic element of the π-system can be written as $F_i = F_{i1} \cap \cdots \cap F_{im_i}$ with $F_j \in F_{ij}$. From here, it becomes clear from the fact that the collection of $\mathcal{F}_{ij}$ ranging over all i and j are independent that we have independence of these π-systems. It then follows that the families $\sigma(\mathcal{P}_1), \ldots, \sigma(\mathcal{P}_n)$ are independent. Clearly, $\mathcal{G}_i = \sigma(\mathcal{P}_i)$.

2. The random variable $Y_i = f_i(X_{i1}, \ldots, X_{im_i})$ as σ-algebra $\sigma(Y_i)$ generated by inverse images of Borel sets in $\mathbb{R}$ under $(X_{i1}, \ldots, X_{im_i})^{-1} \circ f^{-1}$, which is contained in the σ-algebra generated by the function $(X_{i1}, \ldots, X_{im_i}) : \Omega \to \mathbb{R}^{m_i}$, as f_i is Borel. This σ-algebra is contained in the σ-algebra $\mathcal{G}_i$ generated by each of the individual σ-algebras $\sigma(X_{ij})$ for $j = 1, \ldots, m_i$. Of course, $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are independent from Part A, so these σ-algebras are independent and in turn the $\sigma(Y_i) \subset \mathcal{G}_i$ are independent as well.

Solution II.10 (Problem 6.11)**.** There are two parts:

1. In the case that $X_1, \ldots, X_n$ are independent, then $f_1(X_1), \ldots, f_n(X_n)$ are independent, by the previous problem. It then follows that

$$\prod_i \mathbb{E}[f_i(X_i)] = \prod_i \int_{\mathbb{R}} f_i d\mu_{X_i} = \int_{\mathbb{R}^n} f_1 \cdots f_n d\mu_{X_1} \cdots d\mu_{X_n} = \int_{\mathbb{R}^n} f_1 \cdots f_n d\mu_{\mathbf{X}} = \mathbb{E}[f_1(X_1) \cdots f_n(X_n)] \tag{11}$$

where we are using Problem 6.8 above. Conversely, suppose the product formula holds. In particular, we have, for all bounded and continuous $f_1, \ldots, f_n$:

$$\int_{\mathbb{R}^n} f_1 \cdots f_n d\mu_{X_1} \cdots d\mu_{X_n} = \int_{\mathbb{R}^n} f_1 \cdots f_n d\mu_{\mathbf{X}} \tag{12}$$

We can leverage this formula to show that $\mu_{X_1} \otimes \cdots \otimes \mu_{X_n}$ and $\mu_{\mathbf{X}}$ agree on rectangles. In particular, note that $\mu_{X_j}([a_j, b_j]) = \mu_{\mathbf{X}}(\mathbb{R} \times \cdots \times [a_j, b_j] \times \cdots \times \mathbb{R})$, hence a continuous ε-approximation f_i of the indicator $\chi_{[a_j,b_j]}$ relative to μ_{X_j} in L^1 is a ε-approximation of $\chi_{\mathbb{R}\times\cdots\times[a_j,b_j]\times\cdots\times\mathbb{R}}$ relative to $\mu_{\mathbf{X}}$ in L^1. Assume WLOG that $|f_i| \leq 1$ everywhere, so that $\int_{\mathbb{R}^n} f_1 \cdots f_n d\mu_{\mathbf{X}}$ is $n\varepsilon$-close to $\mu_{\mathbf{X}}([a_1, b_1] \times \cdots \times [a_n, b_n])$ and the product of the integrals over f_j is similarly $n\varepsilon$-close to the product of the $\mu_{X_j}([a_j, b_j])$. In particular,

$$\left| \mu_{\mathbf{X}}([a_1, b_1] \times \cdots \times [a_n, b_n]) - \prod_{j=1}^{n} \mu_{X_j}([a_j, b_j]) \right| < 2\varepsilon n \tag{13}$$

for any $\varepsilon > 0$, hence they are equal, and the measures agree on rectangles. Thus, they agree on the algebra of disjoint unions of rectangles, so they agree on the σ-algebra genrated by this algebra: all Borel sets, therefore are equal. So the random variables are independent.

2. In the case that $X_n^i \to X^i$ almost surely, then $f_i(X_n^i) \to f_i(X^i)$ almost surely for $f_1, \ldots, f_m$ bounded continuous. It then follows from dominated convergence that $\mathbb{E}[f_i(X_n^i)] \to \mathbb{E}[f_i(X^i)]$ for $n \to \infty$. In addition, $\prod_i f_i(X_n^i) \to \prod_i f_i(X^i)$ almost surely and again by dominated convergence $\mathbb{E}\left[\prod_i f_i(X_n^i)\right] \to \mathbb{E}\left[\prod_i f_i(X^i)\right]$ for $n \to \infty$. Because we have $\prod_i \mathbb{E}[f_i(X_n^i)] = \mathbb{E}\left[\prod_i f_i(X_n^i)\right]$ for all n, then we have equality in the limit, and the random variables are independent.

Solution II.11 (Problem 6.12)**.** TODO

Solution II.12 (Problem 6.13)**.** There are three parts:

1. Note that $XY \in L^1$ by Holder's inequality. In addition, $\int_\Omega X d\mathbb{P} \leq \left(\int_\Omega X^2 d\mathbb{P}\right)^{1/2} < \infty$ (again by Holder). The same holds for Y. Thus, $\mathrm{Cov}(X, Y)$ is well-defined and is equal to $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

2. This follows from Proposition 6.17 applied by $X, Y \in L^1$ (as they are in L^2).

3. Let Ω be the probability space corresponding to flipping two coins, define $X(H, H) = 1 = -X(T, T)$, $X(H, T) = 0 = X(T, H)$. Also define $Y(H, H) = 0 = Y(T, T)$ and $Y(H, T) = 1 = -Y(T, H)$. Then clearly $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ and $XY = 0$, so $\mathbb{E}[XY] = 0$. However, X and Y are not independent as $\mathbb{P}[X = 0, Y = 0] = 0$, while $\mathbb{P}[X = 0] = \frac{1}{2}$ and $\mathbb{P}[Y = 0] = \frac{1}{2}$.

Solution II.13 (Problem 6.14)**.**

III. Chapter 7 (Weak Convergence)

Problem III.1 (Problem 7.1)**.** Let F be closed in metric space (S, d), then for $\varepsilon > 0$ there exists a Lipschitz bounded function $f_{F;\varepsilon} : S \to \mathbb{R}$ lying between 0 and 1, is equal to 1 for $x \in F$, and is equal to 0 for $x \in S$ such that $d(x, F) \geq \varepsilon$

Proof. The function $g(x) = d(x, F)$ is Lipschitz. Suppose $d(x, F) = 0$, then we can pick a sequence of points $x_n \in F$ which become arbitrarily close to x in the metric d, so $x \in F$. Thus, given $x, y \in S$, if $x \in F$, then $|d(y, F) - d(x, F)| = d(y, F) \leq d(x, y)$ by definition, so g is Lipschitz in this case. Otherwise, if $x, y \notin S$, so $d(x, F), d(y, F) > 0$. Pick $z \in F$ such that $d(x, z) < d(x, F) + \varepsilon$. Since $d(y, z) \geq d(y, F)$, then

$$d(y, F) - d(x, F) - \varepsilon \leq d(y, z) - d(x, z) \leq d(x, y) \tag{14}$$

for all ε, so $d(y,F) - d(x,F) \leq d(x,y)$. A similar argument but swapping x and y shows that $d(x,F) - d(y,F) \leq d(x,y)$, hence $|g(y) - g(x)| \leq d(x,y)$, as desired. From here, we let $h : [0,\infty) \to [0,1]$ be 0 on $[\varepsilon,\infty)$, 1 at 0, and interpolating linearly between 0 and 1 on $[0,\varepsilon]$. Clearly, $f_{F;\varepsilon} = h \circ g$ has the desired properties. $\square$

IV. Chapter 10 (Conditional expectation)

V. Chapter 11 (Discrete martingales)